MOSAIC: Data-Certified Stochastic Release under Structured Deployment Shift

Rudra Chopra
Independent Researcher
Contra Costa County, California, USA
Code and artifacts: https://github.com/RudraChopra/mosaic-certified-release/tree/main

Abstract

AI systems increasingly expose small decision interfaces, such as routing, moderation, eligibility, or triage signals derived from richer internal representations. We introduce MOSAIC, a software-only release layer that emits a task-specific stochastic token only when its strongest source-inference attacker and worst-stratum task error are both certified under data-supported deployment shift; otherwise it abstains. A simultaneous confidence event is placed on pre-release token laws before channel optimization, so it covers the entire stochastic-channel family searched on that table. Labeled target data then certify a common-transform-plus-residual bridge class. We prove exact finite-sample attacker and utility envelopes, global finite-alphabet optimization, matching universal-region sample-complexity bounds, bounded multi-item transcript accounting, and anytime-valid recertification. In 1,000 locked safety trials, naive continuum selection releases contract-violating channels on 42.1% of tables; MOSAIC records none. The registered five-benchmark study releases 20/100 interfaces, all on BiasBios-Clinical, with 0/20 held-out violations. Five locked natural and temporal-shift confirmations, including Qwen2.5-1.5B hidden states and Camelyon pathology, release 85/130 interfaces with 0/85 held-out and 0/1,000 operational replay violations. Under disclosed post-hoc bridge-admitted stress laws, a direct target-table rule violates 16/36 contracts; MOSAIC retains the 20 safe releases and abstains on all 16 failures.

1 Introduction

Routing, moderation, eligibility, and triage systems often expose a compact decision signal derived from a richer model state. Representation editing is commonly evaluated by fitting a probe to that state (Ravfogel et al. 2020, 2022; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026), although probe design can change the apparent conclusion (Pimentel et al. 2020). This leaves two deployment gaps: a different downstream rule may recover the attribute, and a release selected on a noisy validation table may fail after population shift. The operational object that needs a contract is the public interface, not the claim that every trace of a concept vanished internally.

We study a task-specific finite-token interface. A tokenizer fixed without the certification fold maps an internal representation to $C \in [K]$; a source-blind stochastic channel $M \in \mathrm{Stoch}(K, L)$ releases the only public token Z. Within each task label, the contract bounds the balanced accuracy of the Bayes-optimal source attacker rather than a chosen probe. This isolates conditional source signal from label prevalence. Pre-release shift is a common Markov transform plus bounded source-specific contamination, permitting large source-blind drift while charging the part that can create new source differences.

MOSAIC, Minimax-Optimized Source-Agnostic Invariant Channels, is a general certified-release layer instantiated here for edited representations. It builds an exact attacker envelope from simultaneous multinomial confidence regions. Because the event precedes M, it covers every stochastic channel, including one optimized on the same table. A bridge LP uses labeled target data to learn a common transform and the largest certifiable retained mass; exact transform-and-contamination envelopes then certify source distinguishability and a jointly selected decoder. Finite-alphabet optimization is global, and MOSAIC returns ABSTAIN when no feasible release exists.

The central object is the data-certified bridge: a finite-sample test that turns labeled target observations into a population class on which the chosen interface remains valid. The upstream confidence event then covers the whole post-channel family, so privacy and utility can be optimized jointly without testing each channel. This differs from finite-candidate risk control because the event lies before the channel search. Locked experiments compare this transport certificate with direct target-table certification, plug-in and held-out selection, LTT, deterministic release, official erasers, and an exact population oracle.

2 Related Work

Concept-erasure methods remove source-associated directions, covariance, or components (Ravfogel et al. 2020, 2022; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026), while leakage audits show why a weak probe is not a universal guarantee (Pimentel et al. 2020; Elazar et al. 2021; Chowdhury et al. 2025). FARE certifies every downstream classifier for restricted finite encoders, and Fair Normalizing Flows bounds strong adversaries through tractable densities (Jovanovic et al. 2023; Balunović, Ruoss, and Vechev 2022). Stochastic preprocessing and fair repre-

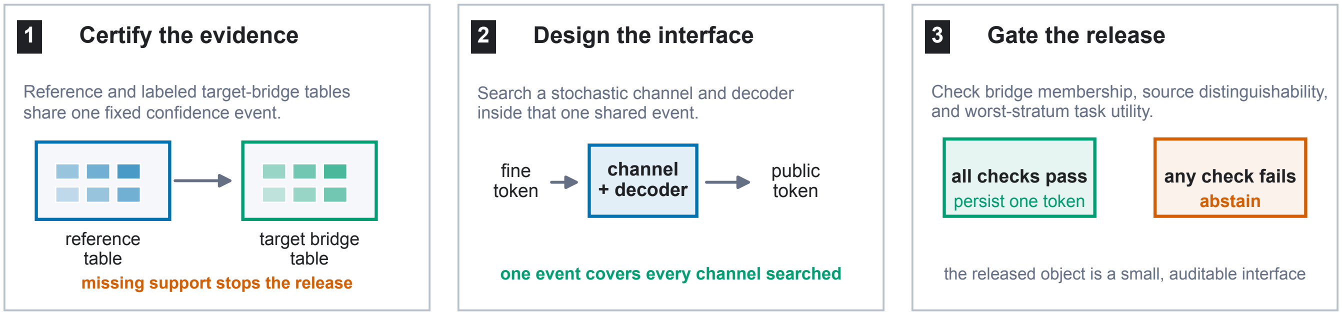

Figure 1: MOSAIC has three steps: certify the reference and labeled target-bridge evidence, design a source-blind stochastic token interface on that shared event, and gate deployment on bridge membership, source distinguishability, and task utility. The runtime persists one token only after every check passes; otherwise it abstains.

sentations motivate randomized release channels (Calmon et al. 2017; Makhdoumi et al. 2014; Agarwal and Deshpande 2024). Related fairness certificates and transfer analyses control different population-level criteria under stated shift assumptions (Gitiaux and Rangwala 2021; Chen et al. 2022; Kang et al. 2022; Agarwal et al. 2025).

LTT, conformal risk control, and Pareto testing certify registered decisions and may abstain (Bates et al. 2021; Angelopoulos et al. 2025, 2024; Laufer-Goldshtein et al. 2023); robust validation addresses declared uncertainty sets (Tibshirani et al. 2019; Cauchois et al. 2024). MOSAIC differs by certifying the categorical experiment before channel search, covering an uncountable channel family without separate tests. Its source-blind transform is a Blackwell garbling and its residual is a contamination neighborhood (Blackwell 1953; Le Cam 1964; Torgersen 1991; Huber 1964). The bridge makes that shift model testable from target data, whereas unrestricted shift is impossible to certify (Ben-David et al. 2010; Klivans, Stavropoulos, and Vasilyan 2024; Chandrasekaran et al. 2025). Selective classification supplies the broader abstention perspective (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2017); group-robust learning targets worst-group prediction rather than certification of a released interface (Sagawa et al. 2020).

3 Problem Setup

Let $S \in [G]$ denote source, $Y \in [J]$ task label, and $C \in [K]$ the fine token. For a fixed label, write $p_{s,y}(c) = \Pr(C = c \mid S = s, Y = y)$. The source-blind release channel M produces $Z \in [L]$. The balanced accuracy of the strongest downstream source attacker within label y is

$$A_y(P, M) = \frac{1}{G} \sum_{z=1}^{L} \max_{s \in [G]} (p_{s,y} M)(z), \tag{1}$$

with normalized advantage $\mathrm{Adv}_G(A) = (GA-1)/(G-1)$. This is chance-normalized and ranges from zero to one. The registered source contract is $\max_y \mathrm{Adv}_G(A_y) \le \tau_P$.

Running case. We follow one registered BiasBios-Clinical interface throughout the paper. Binary gender is the audited source and a clinical-profession decision is the task. After rank-four INLP, a fixed task score maps each biography to one of four fine tokens. MOSAIC releases two tokens with decoder $g(z) = z$ and

$$M = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ .725 & .275 \\ 0 & 1 \end{pmatrix}.$$

It never exposes the edited embedding. The source-advantage bounds are .201 and .204 within the two task labels, and the certified worst-stratum task error is .261, clearing the registered .35 and .40 contracts. The untouched diagnostic values are .052 and .184.

C	fine token	Z	public token
M	channel	g	task decoder
$p_{s,y}$	reference law	$q_{s,y}$	target law
$t_y, T_y, r_{s,y}$	bridge terms	τ_P, τ_U	contracts

The main contract concerns source inference within a task label. A joint-law extension covers natural-mixture source inference: define $X = (Y, C)$, estimate $P(X \mid S)$ without balancing labels, and lift the channel as $\widetilde{M}((y, c), z) = M(c, z)$. The same envelope then certifies inference from Z; using output (Y, Z) covers an attacker given the true label as well.

For each supported source–label stratum, the certification table supplies an empirical law $\widehat{p}_{s,y}$ and radius $\epsilon_{s,y}$. The simultaneous confidence set is $\mathcal{C}_{s,y} = \{p \in \Delta_K : \|p - \widehat{p}_{s,y}\|_1 \le \epsilon_{s,y}\}$. Under conditionally i.i.d. categorical sampling within each registered source–label stratum, the simultaneous event

$$\mathcal{E} = \bigcap_{s,y} \{\|\widehat{p}_{s,y} - p_{s,y}\|_1 \le \epsilon_{s,y}\} \tag{2}$$

has probability at least $1 - \delta$ after a fixed allocation across strata. We use the Weissman multinomial radius (Weissman et al. 2003) in the confirmation, although any valid simultaneous region can replace it. The tokenizer and fine alphabet must be fixed without these observations.

The external law for label y belongs to the structured class

$$q_s = t_y p_s T_y + (1 - t_y) r_s, \qquad t_y \ge 1 - \eta_y, \tag{3}$$

where T_y is a source-independent fine-token channel from a registered convex uncertainty set and each residual r_s is

Framework	Certified object	Difference from MOSAIC
FARE (Jovanovic et al. 2023)	Downstream rules for a fixed finite encoder	No same-table stochastic-channel search or target-shift certificate
LTT and risk control (Angelopoulos et al. 2025; Bates et al. 2021)	Risks of registered candidates	Candidate-level tests rather than one pre-channel event for the continuum
Blackwell and deficiency (Blackwell 1953; Le Cam 1964)	Population comparisons between experiments	Shift structure is assumed rather than certified from finite target data
Robust and testable shift (Cauchois et al. 2024; Klivans, Stavropoulos, and Vasilyan 2024)	Predictor risk or a shift hypothesis	No joint optimization of an exact source attacker and public token
MOSAIC	Source attacker and task risk for an optimized finite token	Pre-channel coverage plus a data-certified target-shift bridge

Table 1: Closest certification frameworks by object and scope. MOSAIC combines one pre-channel confidence event, an exact Bayes-attacker envelope, and finite-sample evidence for the target-shift class.

arbitrary. Both components pass through the same release channel M. This placement matters: a constant-row software channel cannot leak information that appears before it.

4 Adaptive Within-Label Attacker Certificate

For a bounded vector v, define the exact confidence-set support function

$$\Phi(\widehat{p}, \epsilon; v) = \max_{p \in \Delta_K : \|p - \widehat{p}\|_1 \le \epsilon} p^\top v. \tag{4}$$

It is obtained by moving at most $\epsilon/2$ mass from low to high coordinates of v, or from an equivalent linear-program dual. For attacker assignment $a \in [G]^L$, let $v_{M,a,s}(c) = \sum_z M(c,z)\mathbf{1}\{a(z) = s\}$. We then define

$$\overline{A}(\widehat{P}, M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \Phi(\widehat{p}_s, \epsilon_s; v_{M,a,s}). \tag{5}$$

Theorem 1 (Adaptive exact envelope)**.** *On $\mathcal{E}$, simultaneously for every row-stochastic M, including a channel selected as an arbitrary function of the same table, $A(P, M) \le \overline{A}(\widehat{P}, M)$. For fixed M, the right-hand side is the exact supremum over the product of the stated ℓ_1 confidence sets.*

The proof writes the Bayes attacker as a maximum over deterministic assignments. For a fixed assignment the source confidence sets separate, giving one support function per source; all maxima are finite and commute. The implication is pointwise over the complete stochastic-channel space, which is why adaptive selection needs no channel-count correction. In the running case, the same confidence event covers the displayed M and every alternative four-by-two channel considered by the optimizer.

5 Data-Certified Shift Membership

MOSAIC certifies structured-shift membership from labeled target bridge data. Let a labeled bridge sample from the target population define simultaneous confidence sets $\mathcal{D}_{s,y}$ for its fine-token laws $q_{s,y}$. Write

$$h_{s,y}(w) = \max_{p \in \mathcal{C}_{s,y}} p^\top w, \qquad \underline{q}_{s,y}(z) = \min_{q \in \mathcal{D}_{s,y}} q(z). \tag{6}$$

For each label, the bridge program chooses retained mass t and $W = tT$ by solving

$$\begin{aligned} \max_{t,W} \quad & t \\ \text{s.t.} \quad & W \ge 0, \quad W\mathbf{1} = t\mathbf{1}, \quad 0 \le t \le 1, \\ & h_{s,y}(W_{:z}) \le \underline{q}_{s,y}(z) \quad \forall s, z. \end{aligned} \tag{7}$$

The support functions and coordinate lower bounds have exact LP duals, so Eq. (7) is one linear program rather than a transform search.

Running case, continued. For the released BiasBios interface, the bridge LP certifies retained common mass .799 and .796 in the two task labels. MOSAIC therefore charges source-specific residuals of .201 and .204 in its downstream attacker and utility envelopes. These are population bounds from separate reference and bridge confidence regions, not fitted similarities on the diagnostic fold. The resulting channel merges the first two score bins, randomizes only the third, and preserves the fourth: this is the complete public decision surface an auditor must inspect.

Theorem 2 (Adaptive finite-sample bridge)**.** *Suppose the reference and bridge confidence sets jointly cover every $p_{s,y}$ and $q_{s,y}$. Any feasible $(\widehat{t}, \widehat{W})$ with $\widehat{t} > 0$, including the same-table maximizer, defines $\widehat{T} = \widehat{W}/\widehat{t}$ and residual laws satisfying*

$$q_{s,y} = \widehat{t}\, p_{s,y} \widehat{T} + (1 - \widehat{t}) r_{s,y} \quad \forall s. \tag{8}$$

Moreover, Eq. (7) *returns the largest retained mass certified uniformly over the product confidence region.*

Indeed, feasibility and coverage imply $q_{s,y}(z) \ge p_{s,y}^\top \widehat{W}_{:z}$ coordinatewise. The difference is nonnegative and sums to $1 - \widehat{t}$, so normalization constructs $r_{s,y}$. Conversely, robust membership requires exactly these coordinatewise inequalities; maximizing the reference support and minimizing the bridge coordinate yields Eq. (7). Thus the learned singleton transform $\{\widehat{T}_y\}$ and contamination $\widehat{\eta}_y = 1 - \widehat{t}_y$ can be inserted into the certificates below on the same event. If a required bridge source–label stratum is absent, its confidence set is the full simplex, every coordinate lower bound is zero, and Eq. (7) forces $\widehat{t} = 0$. For the practitioner, the bridge either supplies a measured residual budget for the target population or stops the release before a channel is deployed.

6 Certification under Pre-Release Shift

We use three names consistently: the *data-certified bridge* tests shift membership, the *transform-exact certificate* enumerates the learned shift geometry, and the *capacity fallback* upper-bounds it when that geometry cannot be enumerated.

Define the arbitrary differential-shift capacity

$$\mathrm{Cap}_G(M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \max_c v_{M,a,s}(c). \tag{9}$$

This equals $\sup_{r_1,\dots,r_G} A(R, M)$. For $G = 2$, its normalized value is the Dobrushin coefficient, the largest total variation distance between two rows of M.

For a transform polytope with registered extremes $\mathcal{V}_y = \mathrm{ext}(\mathcal{T}_y)$, define the shifted envelope

$$\overline{A}_y^X(M) = \frac{1}{G} \max_{T \in \mathcal{V}_y,\, a \in [G]^L} \sum_s \Big[(1 - \eta_y)\Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; T v_{M,a,s}) + \eta_y \max_c v_{M,a,s}(c) \Big]. \tag{10}$$

Theorem 3 (Transform-exact structured-shift certificate)**.** *On $\mathcal{E}$, every same-table selected M and every external law in Eq.* (3) *satisfy $A_y(Q, M) \le \overline{A}_y^X(M)$. For fixed M, the right-hand side is the exact supremum over the stated confidence sets, transform polytope, retained masses, and arbitrary residual laws.*

For fixed T, retained mass, and attacker assignment, the source confidence sets and residual simplexes separate. Their support values are respectively $\Phi(\widehat{p}, \epsilon; Tv)$ and $\max_c v(c)$. The latter is no smaller, so the worst retained mass is $1 - \eta_y$. The support objective is convex in T; its maximum over a polytope occurs at an extreme. These finite maxima commute, proving exactness. Since the statement holds pointwise for every M, it survives adaptive channel selection.

When transform extremes cannot be enumerated, a capacity-fallback corollary is available. Define $\rho_T(M) = \max_c \mathrm{TV}((TM)(c, \cdot), M(c, \cdot))$ and $B_y(M) = \min\{\mathrm{Cap}_G(M), \overline{A}_y(\widehat{P}, M) + \rho_{\mathcal{T}_y}(M)\}$. Then

$$A(Q, M) \le (1 - \eta_y) B_y(M) + \eta_y \,\mathrm{Cap}_G(M). \tag{11}$$

The exact envelope in Eq. (10) is never larger. When $\eta_y = 0$ and $TM = M$, both reduce to the reference certificate; when $\eta_y = 1$, both reduce to exact distribution-free capacity.

Because the fallback separately upper-bounds every law in the same structured class, both transform-exact source and utility values are pointwise no larger than their fallback values. "Never worse" is therefore an implementation check implied by the theorem; experiments measure how often tightness changes the release decision. Proofs are in the supplement.

For decoder $g : [L] \to [J]$, let $u_y(c) = \sum_z M(c, z)\mathbf{1}\{g(z) \ne y\}$. The matching exact conditional error certificate is

$$\overline{E}_{s,y}^X(M, g) = \max_{T \in \mathcal{V}_y} \Big[(1 - \eta_y) \Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; T u_y) + \eta_y \max_c u_y(c) \Big]. \tag{12}$$

It is the exact supremum over the same confidence and shift sets. Thus M, g, within-label source distinguishability, and worst-stratum utility can all be selected on one event.

Guarantee at a glance. With probability at least $1 - \delta$ over the registered reference and bridge samples, every released (M, g) meets both contracts for every target law in the learned bridge class. If the bridge has zero retained mass, a required stratum is missing, or no channel satisfies both bounds, MOSAIC abstains.

Statistical and session extensions. For a distribution-uniform ℓ_1 region that must cover every bounded functional selected from the same K-category table, the minimax resolution rate is $n = \Theta((K + \log(1/\delta))/\gamma^2)$; the Weissman construction matches it up to constants (Han, Jiao, and Weissman 2015). For a registered session of r possibly correlated fine items, compiling the mechanisms into the joint transcript channel gives the same exact envelope. A scalable capacity bound is $1 - \prod_{i=1}^{r}(1 - \alpha_i)$, where α_i is the Dobrushin coefficient of item channel i. Finally, replacing fixed-time regions by Dirichlet-mixture e-process regions makes every recertification time and optional stopping rule valid on one event (Ryu and Wornell 2024). The supplement proves these statements, gives a finite-cover extension for continuous release classes, and an all-threshold DKW corollary.

Unrestricted source-specific shift can be erased only by a fine-token-independent channel, so a bounded residual is a deployment assumption rather than a hidden technical convenience. Likewise, a missing required source stratum is unestimable rather than evidence of safety; both boundary proofs are in the supplement. A constant-row channel is the $\varepsilon = 0$ endpoint of local differential privacy (Warner 1965).

7 Global Optimization and Abstention

For fixed decoder, support functions, transformed scores, and residual maxima have linear epigraphs. Enforcing every transform extreme and attacker assignment gives one LP in the KL channel entries; enumerating J^L decoders yields the global finite-alphabet optimum. Acceptance requires solver optimality, valid primal residuals, and independent certificate recomputation.

Release protocol. Fix tokenizers, alphabets, thresholds, split roles, and confidence allocation before viewing certification or bridge folds. Construct simultaneous regions, solve Eq. (7), and enumerate decoder LPs subject to every source constraint. Replay each returned certificate, then select the feasible registered tokenizer with minimum certified error under a fixed tie break. Zero retained mass, missing support, no feasible row, or a failed numerical check returns ABSTAIN; otherwise only (M, g) and Z are released.

With $R = \max_y |\mathcal{V}_y|$, the direct formulation is polynomial in K for fixed (G, J, L, R) and exponential only in the released alphabet L. Constraint generation adds the strongest omitted attacker until none violates the contract; the supplement gives the proof and scaling details.

8 Evaluation Design

Study map. The evaluation separates four questions: whether certification blocks unsafe selection, whether bridge membership is testable, whether useful interfaces remain releasable, and whether the result transfers across domains. Protocols fix data roles, thresholds, seeds, and failure gates before outcomes. Diagnostic folds never enter channel or candidate selection.

Synthetic safety and bridge power. The safety study uses two source–label populations, two shift regimes, eight rules, up to five sample sizes, and 1,000 independent tables per cell at familywise $\delta = .05$. A separate population evaluator grades every selected channel. Comparators include held-out certification, finite-family LTT, and deterministic release. A second 12,000-table study varies bridge support from 500 to 5,000 observations per source–label stratum. It compares a valid common transform with source-specific transforms and contamination above the declared budget, using a retained-mass threshold of .80.

Five-benchmark release study. We registered 100 fresh jobs: 20 untouched seeds on each of Waterbirds, Camelyon17-WILDS, CivilComments-WILDS, BiasBios-Clinical, and GaitPDB, spanning vision, medical vision, text, and time series (Koh et al. 2021). Each job evaluates 13 frozen feature edits from the official INLP, LEACE, R-LACE, TaCo, and MANCE++ implementations; proxies are forbidden. Construction data alone fit a balanced task score and its four-bin tokenizer. Separate reference observations certify the pre-shift table, while the natural external split is divided within each supported source–label stratum into two-thirds labeled bridge data and an untouched one-third diagnostic fold. Neither diagnostics nor their outcomes enter channel or candidate selection.

The familywise level is .05 over 13 candidate tables and both reference and bridge regions. The source-advantage contract is .35; worst-stratum utility thresholds are .30, .35, .40, .45, and .49, with .40 primary. We globally optimize $L = 2$, then reoptimize $L = 4$ only for the best covered $L = 2$ candidate. Locked comparators use the same candidates and folds: capacity transfer, a bridge plug-in without intervals, a validation plug-in that ignores the bridge, and unconditional deployment of the best validation plug-in. All datasets, thresholds, abstentions, and missing-support outcomes remain reportable.

Target labels and audit controls. Target S and Y labels supply the conditional token laws $q_{s,y}$ used by the bridge and within-label contract. Where S is sensitive, collection, consent, and permitted audit use are part of the deployment protocol; unsupported strata trigger abstention. Strict numerical replay and an exact-rational audit recheck every release decision. The supplement preserves the disclosed structural-zero repair, earlier artifacts, and complete amendment history.

Geographic-shift confirmation. Before downloading the target states, we register 60 Folktables jobs (Ding et al. 2021): three binary tasks (income, employment, and public coverage), four natural state shifts from California to Washington, Illinois, New York, and Florida, and five seeds. Binary sex is the audit source and is excluded from every released feature matrix. California supplies eraser-train and reference folds. Within each target state, a seeded partition of complete Public Use Microdata Areas supplies the labeled bridge and untouched diagnostic folds; no person or PUMA crosses that boundary. Every job evaluates the same 13-candidate official frontier at $K \in \{4, 8\}$, five registered utility thresholds, and 100 one-token operational draws per release. We hash-lock all 12 raw-data manifests and processed stores before any outcome is computed.

Image-origin confirmation. CINIC-10 provides file-level CIFAR-10 versus ImageNet origin (Darlow et al. 2018). A locked five-seed study uses ImageNet-pretrained ResNet-18 features, a vehicle-versus-animal task, and disjoint construction, reference, bridge, and diagnostic folds. Each seed compares the unedited interface with official closed-form LEACE under the same .35 source and .40 utility contracts; neither origin labels nor embeddings are released. A disclosed sequential extension reruns 40 fixed seeds under one stricter familywise allocation.

9 Results

Synthetic selection safety. All preregistered gates pass. Figure 2 reports every implementable rule in the primary cells; "capacity fallback" is MOSAIC's generic capacity certificate. Independent replay reconstructs every decision, population risk, and interval with zero mismatch.

The paired safety difference is 42.1 points (95% bootstrap interval $[39.0, 45.2]$), with 421 plug-in-only violations and no reverse discordance. Across 8,000 capacity-fallback certification tables, no accepted release violates the exact population contract. In the retention cell, capacity-fallback safe deployment exceeds held-out, finite-LTT, and deterministic rates by 53.8, 53.0, and 56.3 points. No deterministic channel is feasible. These finite-run facts align with the $1 - \delta$ guarantee.

A separately locked 5,000-table refinement evaluates exact shift geometry on the same tables. At $n = 125$, transform-exact certification safely deploys on 53.0% versus 0.3% for fallback; at $n = 250$, the rates are 99.9% and 57.8%. The exact objective is no larger on all 5,000 pairs, as theory requires, and all 10,000 method-cell decisions have zero false acceptances. Full cell tables, threshold sensitivity, amendment history, and audits are in the supplement.

Matched finite-family baselines. A separately locked replay places four rules on the same 1,000 tables in each primary scenario. In the retention cell, continuum MOSAIC certifies 579/1,000 deployments (57.9%, exact 95% interval $[54.77, 60.98]$), versus 283/1,000 (28.3%, $[25.53, 31.20]$) for Holm LTT over all 500 fixed channel–decoder pairs. MOSAIC alone deploys on 312 paired tables, LTT alone on 16, and both on 267, a 29.6-point paired difference. The exact McNemar value is 2.28×10^{-72}; it is reported as a descriptive paired statistic. The confidence-region grid and

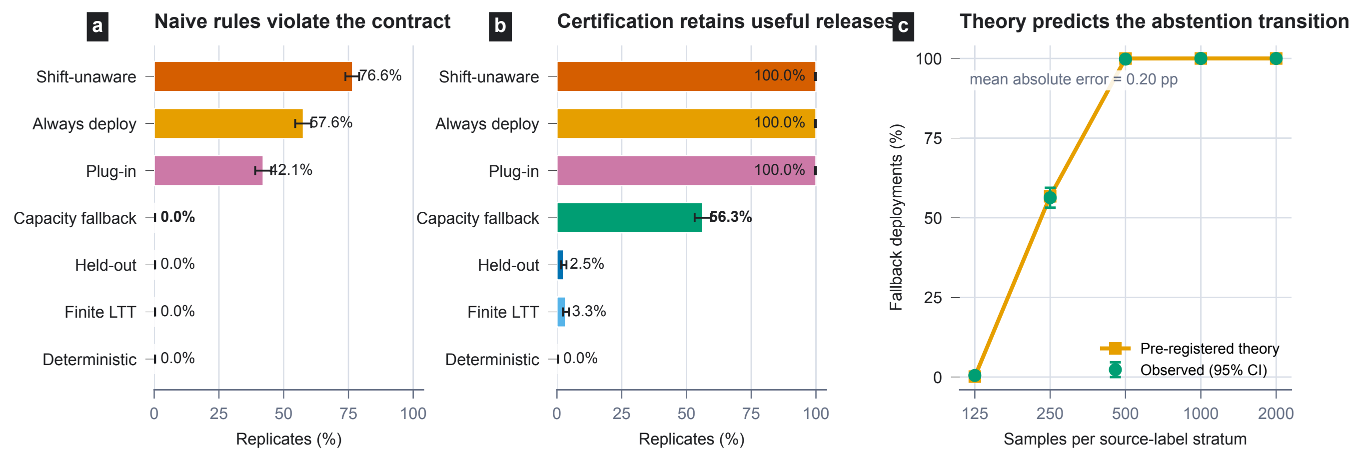

Figure 2: Hash-locked confirmation with 1,000 independent certification tables per cell. Error bars are exact 95% Clopper–Pearson intervals. (a) At the safety boundary, plug-in continuum selection falsely accepts 42.1% of tables; MOSAIC abstains with no false acceptance. Ignoring shift is worse. (b) In the retention cell, the capacity fallback safely deploys on 56.3%, versus 2.5% for held-out certification, 3.3% for finite LTT, and 0% for deterministic MOSAIC. (c) Deployment follows the preregistered theory curve with 0.20 percentage-point mean absolute error. Exact 95% Clopper–Pearson intervals are shown.

matched deterministic restricted-encoder adaptation both abstain throughout this cell. All four rules have zero observed false acceptance. We do not label the deterministic row as FARE: official FARE trains a fair decision tree for demographic parity on its own raw-input protocol, whereas this study asks about within-label source inference under a pre-release geographic-shift contract. The row isolates the shared restricted- encoder idea without presenting a non-commensurate end-to-end comparison.

Bridge validity and power. Across a separate 12,000-table locked stress study, a correctly specified common transform is accepted on 0%, 31%, 100%, and 100% of tables as the bridge stratum size rises through 500, 1,000, 2,000, and 5,000. Underdeclared contamination has population retained mass $.775 < .80$; a source-specific transform has minimum retained mass .50. Both violating populations are rejected on every table at every sample size. Independent replay finds no failure on the joint confidence event and no invalid false acceptance. The study establishes power against both registered bridge violations and shows the acceptance transition as bridge support grows.

Five-benchmark release ladder. At the primary $\tau_U = .40$ error contract, equivalently a .60 worst-stratum accuracy floor, strict MOSAIC deploys on 20/100 jobs (20.0%, exact 95% interval $[12.67, 29.18]$), all externally estimable, with 0/20 observed violations (one-sided 95% upper bound 13.91%). The selected official edits comprise six INLP, five LEACE, and nine MANCE++ candidates.

Moving down Table 2 removes one safeguard at a time. Capacity transfer replaces the learned geometry with a worst-case charge; the bridge plug-in removes confidence regions; validation-only selection removes the target-membership gate; and unconditional deployment removes abstention. The resulting changes in release and violation counts isolate the role of each component.

Rule	Released	Unsupported	Diagnostic violations
MOSAIC	20/100	0	0/20
Capacity transfer	0/100	0	–
Bridge plug-in	47/100	0	7/47 (14.9%)
Validation only	80/100	20	18/60 (30.0%)
Always deploy	100/100	20	38/80 (47.5%)

Table 2: Primary five-benchmark ladder at $\tau_U = .40$. Unsupported jobs lack a required target source stratum and are excluded only from the diagnostic violation denominator.

MOSAIC retains 20 safe deployments at every threshold .30–.45 and adds three safe Waterbirds deployments at .49. The primary releases are all on BiasBios-Clinical, where certified median retained mass is .797. Missing source support or an infeasible joint source–utility contract causes abstention on the other benchmarks. In particular, the Waterbirds bridge plug-in releases all 20 jobs, while strict MOSAIC declines because the joint contract is not certified.

Concrete released interface. One BiasBios-Clinical seed selects INLP at rank four for the binary clinical profession task, with binary gender used only as the audited source. A fixed task score bins each biography into four fine tokens; MOSAIC then emits one of two persistent stochastic public tokens rather than the edited embedding or the source label. Its certified source-advantage upper bound is .204 and its certified worst-stratum error upper bound is .261, below the registered .35 and .40 contracts. On the untouched diagnostic fold, the corresponding values are .052 and .184. This is the deployed object: a small versioned routing interface with a decoder, not a claim that the underlying representation is universally erased.

Direct target-table comparator. An audited comparator applies the same simultaneous envelope directly to the labeled target bridge table, without making a bridge transport

Target-law scope	Direct table	MOSAIC
Bridge-admitted, outside direct region (36 real jobs)	16 violations	20 releases; 0 violations
Direct-region, bridge-invalid (7,983 locked tables)	region contains q	7,983 abstentions

Table 3: Complementary scope cells. The first is the post-hoc real-table stress; the second reuses two prelocked bridge-invalid target laws.

claim. It deploys 36/100 jobs and records 0/36 diagnostic violations, versus 20/100 and 0/20 for strict MOSAIC. This retention gain disappears once the target law moves inside the learned bridge class but outside the sampled-table region. In this post-hoc stress test, for each of the 36 direct releases, we construct the bridge-admitted law that maximizes its source attacker. The direct rule then violates 16/36 contracts (44.4%, exact 95% interval $[27.9, 61.9]$); MOSAIC releases the other 20, violates none, and abstains on all 16 failures. Every constructed law is an exact member of its stored bridge class and lies outside the direct confidence region. Median structural residual radius is $.279$, and median distance beyond the direct region is $.368$ in ℓ_1 after deterministic resolution of an exact attacker tie. The supplement gives the construction, numerical interface utility, and threshold frontier.

The second row closes the scope check without claiming a failure of direct target-table certification. In the separate 12,000-table study, the two prelocked violations have population retained mass below $.80$. Their true target laws fall inside the direct target region on 7,983/8,000 invalid tables; MOSAIC's membership gate abstains on all 8,000. Thus direct-region coverage and bridge membership answer different questions, while the bridge has an explicit rejection path when its declared structure fails. The supplement reports the rowwise construction and the full frozen receipt.

Numerical verification and fresh confirmations. Strict-v2 deterministic replay matches all 100 receipts, while exact-rational checks verify 1,300 bridge certificates and 1,400 outward release bounds. A pre-outcome five-seed BiasBios rerun releases every job without a diagnostic violation. The supplement records the independent ACS confirmation, correction history, and decomposed bounds.

Natural multi-state geographic transfer. Before downloading the four target states, we locked 60 ACS jobs spanning income, employment, and public-coverage tasks; Washington, Illinois, New York, and Florida targets; five seeds; and whole-PUMA holdouts. Every job runs the same 13-candidate official frontier at $K \in \{4, 8\}$. At the primary $.40$ contract, $K = 4$ MOSAIC releases 15/60 with 0/15 held-out violations and 0/1,500 violating operational batches. The releases span all four targets (FL=4, IL=2, NY=5, WA=4) and two tasks: employment=13 and income=2. The direct table releases 47/60 with 0/47 held-out violations. At $K = 8$, MOSAIC abstains on all 60 jobs while the direct table releases 47/60 with 0/47 held-out violations, exposing the confidence cost of the larger token table. The final column of Table 4 repeats the held-out diagnostic 100 times, sampling one certified public token per person in each replay.

K	Rule	Releases	Diagnostic viol.	Batch viol.
4	MOSAIC	15/60	0/15	0/1500
4	Direct table	47/60	0/47	2/4700
8	MOSAIC	0/60	–	–
8	Direct table	47/60	0/47	0/4700

Table 4: Fresh natural geographic shift across 60 registered ACS jobs (three tasks, four target states, five seeds). Diagnostic violations are among released jobs with complete held-out PUMA support. Batch violations count 100 operational batch replays per primary release, each sampling one public token per held-out person.

Alphabet sensitivity. Four released tokens improve only five of 100 jobs within the registered 10^{-7} solver criterion, so the real gain comes from certifying shift geometry rather than enlarging the public alphabet. Across 75 synthetic jobs with $K \in \{4, 8, 16, 32, 64\}$ and $G \in \{2, 3, 4\}$, every $.40$ contract certifies; the largest cell has $.237$ median utility slack and $.360$-second median solve time. In the original real ACS $K = 8$ extension, retained mass is $.377$–$.540$, every candidate needs a $.50$-error constant release, and MOSAIC abstains. With a locked increase from 6,000 to 16,000 certification rows per source–label stratum on three employment transfers, $K = 8$ MOSAIC releases Florida and New York with 0/2 held-out and 0/200 operational violations; Illinois abstains. The supplement gives the complete support study.

Natural and temporal evidence. Locked text, census, image, Qwen hidden-state, and Camelyon pathology studies release 85/130 interfaces with 0/85 held-out violations. Qwen and Camelyon each release 5/5, with 0/1,000 operational violations; Camelyon clears even the $.35$ utility contract. A separately frozen 2023 ACS test follows two direct releases first identified in 2021; both violate the $.40$ utility threshold empirically, and one is familywise confirmed (error $.425$, lower bound $.405$). MOSAIC had rejected both.

Power and alternatives. A locked real-proxy study imputes source at $.841$ balanced accuracy, then certifies a non-constant release with source bounds $.218/.279$ and worst-error bound $.369$; smaller calibration folds abstain. Across 35 primary BiasBios and ACS releases, residual shift exceeds sampling uncertainty in every certificate. A matched $\epsilon_{\mathrm{LDP}} = .731$ randomized-response baseline releases 0/35 interfaces at the $.40$ utility contract; MOSAIC releases 35/35 and reduces certified error by a median $.155$. Full curves and costs are in the supplement.

10 Discussion

MOSAIC certifies the categorical experiment before release, covering every same-table M and g. Missing support or expired evidence returns ABSTAIN.

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